



EXPERIMENT NO 3

SLIP TEST ON 3-PHASE SALIENT POLE SYNCHRONOUS MACHINE

AIM

- To conduct the slip test on 3-phase salient pole synchronous machine
- To determine the direct axis and quadrature axis synchronous reactance
- To predetermine the voltage regulation at different loads and power factors and
- To draw the power Vs torque angle characteristics for a specified induced emf

APPARATUS

S.No.	Name of the apparatus	Type	Range	Quantity
1	Voltmeter			
2	Voltmeter			
3	Voltmeter			
4	Ammeter			
5	Ammeter			
6	Rheostat			
7	Tachometer			

PRINCIPLE

The direct and quadrature axis reactance can be measured by slip test. The machine is driven by a dc motor at a speed slightly less or slightly more than synchronous speed. The field winding is kept open circuited and a low voltage 3 phase supply (about 25% of the rated voltage) is applied to the armature terminals. The direction of rotation should be same as the direction of rotating field. If this condition is fulfilled, a small ac voltage would be indicated by the voltmeter across the field winding. The relative velocity between armature mmf and field poles is equal to slip speed i.e. difference between synchronous speed and rotor speed. The stator mmf moves slowly past the field poles at slip speed. This would cause the armature current to vary cyclically at twice the slip frequency. When the peak of the armature mmf is in line with the field poles, the reluctance offered by the magnetic circuit is minimum, the armature current, required for the establishment of constant air-gap flux, will be